

A communication subspace relays context-dependent actions from human prefrontal to motor cortex

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Adaptive behavior relies on the ability to translate abstract rules and goals into actions suited to the current context. Neural population activity in the prefrontal cortex (PFC) has been proposed to support such flexible computations through high-dimensional dynamics, whereas activity in the primary motor cortex (M1) is related more directly to movement execution. How contextual representations in PFC are transformed into ensuing action plans within M1 remains unknown. Previous work suggests that low-dimensional coding subspaces might organize interareal communication, but direct evidence for such population-level communication mechanisms in humans is lacking. Here we use intracranial recordings from human PFC and M1 to identify a communication subspace embedded within high-dimensional PFC activity, that selectively relays behaviorally relevant information at the single-trial level. Activity in this subspace predicts context-dependent action more strongly than either region, revealing a fundamental coding principle by which coordinated interareal population dynamics filter and relay predictive information to guide context-dependent actions.

Context-dependent behavior is a hallmark of human cognitive flexibility, allowing actions to be selected based on current rules, task demands, long-term goals and past experiences^{1–3}. At the core of this adaptability lies predictive processing, which integrates exogenous and endogenous priors to anticipate relevant outcomes, shape sensory processing and optimize action selection^{4,5}. Although distributed networks in the PFC provide the structural basis for these neural computations^{6–8}, the functional mechanisms governing their adaptive processing remain poorly understood. A central challenge in identifying the neural correlates of such complex human behavior is the high-dimensional nature of neurophysiological activity, particularly in association areas such as the PFC^{2,9–12}. This high-dimensionality reflects directly the complex, multiplexed processes that continuously integrate sensory information to drive

context-dependent action execution^{1,3,13}. Mechanisms such as recurrent connections^{3,14}, interareal connectivity^{6,15} and mixed selectivity^{11,12,16}, among other presumed mechanisms, support this high-dimensional nature and render it difficult to isolate behaviorally relevant signatures from population activity. In contrast to complex PFC dynamics, population activity in motor cortex (M1) often unfolds along well-structured trajectories^{17–20}. The PFC–M1 network is thought to support adaptive behavior by routing relevant information dynamically from PFC to M1 (refs. [15,21,22](#)). Consistent with the presumed directional communication pathway, previous studies demonstrated that sensory- and context-related activity emerges first in the PFC and then in M1 (refs. [15,21](#)). However, it remains puzzling how multifaceted representations of contextual information in PFC are transformed into ensuing motor plans.

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Growing evidence suggests that interareal communication is structured through low-dimensional coding subspaces^{23–26}. This concept aligns with the population doctrine, which posits that cell assemblies and their geometric configuration, not individual neurons, constitute the central computational unit of the brain^{27–29}. However, direct experimental evidence for such communication subspaces (CSs) remains sparse^{24–26,30} and is confined largely to studies in nonhuman primates and mice³¹. Although coding subspaces have been described in humans at the level of aggregated single unit^{32,33} and field potentials^{34–36}, it remains unclear whether such low-dimensional subspaces also support PFC–M1 communication in humans and, critically, whether they have a role in higher-order cognition and goal-directed behavior.

Here we determined how context-dependent action information is encoded in the PFC and transformed into motor output to support efficient visual target detection. We hypothesized that PFC and M1 share a joint subspace, analogous to previously studied CSs observed between early and intermediate visual system. Specifically, we reasoned that this subspace might act as a filter for selectively relaying contextually driven action signals from PFC to M1, thereby channeling PFC activity efficiently to downstream motor circuits. If true, then neural activity within this CS should correlate better with context-dependent action than activity within either region alone. The identification of such subspaces requires sufficient spatiotemporal sampling of the underlying neural population. As previous studies relied on aggregated unit firing, we leveraged high-frequency activity (HFA)—a well-established proxy for coordinated population activity³⁷—recorded from densely sampled subdural grid electrodes (electrocorticography, ECoG) covering large portions of PFC and M1 (refs. 38,39). Our findings uncover a shared PFC–M1 CS in which coordinated low-dimensional activity structures information flow from PFC to M1 to enable context-guided target detection, thereby establishing a fundamental principle of neural communication at the population level that supports efficient information transfer in support of goal-directed actions.

Results

To test whether a low-dimensional PFC–M1 CS acts as a selective routing mechanism for filtering context-dependent action information within PFC to M1, we recorded subdural ECoG signals using grid electrodes from 12 pharmacoresistant epilepsy patients (age 40.91 ± 12.32 years, mean $\pm$ s.d.; four female, one unknown) who performed a context-guided target detection task^{40,41}. Participants viewed a continuous stream of triangles with varying orientations and were instructed to respond as quickly as possible when a downward facing triangle (target stimulus (T)) appeared (see Supplementary Table 1 for additional details about the task). Targets were either presented at random or preceded by a fully predictive sequence (100% validity, left-, up- and then right-ward facing triangles; Fig. 1a). Reaction times (RTs) were significantly faster for predicted targets compared to random targets ($P = 0.0333$; $t_{11} = -2.43$, $d = -0.70$; Fig. 1b), indicating that participants relied on the predictive temporal context for effective target detection. To investigate how the prefrontal motor network encodes and transforms this predictive information to guide behavior, we analyzed the simultaneously recorded ECoG signals (PFC: 74 ± 29 contacts, mean $\pm$ s.e.m., M1: 26 ± 7 ; Fig. 1c), using the bipolar referencing scheme. We leveraged specifically ECoG recordings, which provide dense spatial sampling of both regions, to test whether behaviorally relevant information is relayed through structured low-dimensional codes. We hypothesized that a PFC–M1 CS could serve as a filter mechanism to funnel behaviorally relevant information nested in high-dimensional prefrontal dynamics to the lower dimensional motor cortex (Fig. 1d).

Target-related HFA in human PFC and motor cortex

To uncover the neural correlates underlying the behavioral benefit afforded by predicted targets, we analyzed the simultaneously recorded neural activity from the human PFC and M1 for 11 participants (one

patient had to be excluded for excessive high-frequency artifacts). We employed a bipolar referencing scheme along all cardinal and diagonal directions of the ECoG grid (typical dimensions: 8×8 electrodes) but included only unique pairs that could be assigned unambiguously to either the PFC or M1. We quantified HFA as a proxy for coordinated population activity, time-locked to target presentation ($t = 0$). Although we observed a clear increase in target-related activity, there were no significant differences in the overall activity level between predicted and random targets in PFC and M1 (Fig. 2a; cluster-based permutation test for paired samples across timepoints, corrected for multiple comparisons, no clusters were found at $p < 0.05$). However, we did observe that HFA peaks for the predictive trials occurred earlier in the PFC compared to in the random trials (Wilcoxon signed-rank test; $P = 0.0003$) but this effect was not seen in M1 (Wilcoxon signed-rank test; $P = 0.3468$; Fig. 2a). To test whether these context-dependent temporal dynamics were linked to behavior, we examined whether peak latency difference of context-specific target-related activity correlated with the individual behavioral benefit of the predictive relative to the random context (Fig. 2a) but observed no significant correlations in either region (Spearman rank-order correlation; PFC: $r = 0.39$, $P = 0.2345$; M1: $r = 0.35$, $P = 0.2847$).

High-dimensional neural dynamics in human PFC

Although clear target-related activity was evident in both PFC and M1 (Fig. 2a), neither overall activity nor peak latency differences correlated significantly with the prediction effect (Fig. 1b). We hypothesized that this absence was the direct consequence of the high-dimensional nature of prefrontal neural population activity that reflects simultaneously both task-relevant and task-irrelevant features, potentially masking behaviorally relevant activity. Consequently, we considered that context-specific action information might reside only in a subspace of the overall neural population activity (Fig. 1d). To test this hypothesis, we first estimated the intrinsic dimensionality of neural activity in human PFC and M1 using factor analysis⁴² (Methods). At the single-participant level, more factor analysis components were required to explain the full variance in PFC as compared to M1, indicating higher complexity of PFC population activity (Fig. 2b, left). The difference was also evident at the group level, with significantly higher dimensionality in PFC compared to M1 ($P = 0.0174$, $t_{10} = 2.85$, $d = 0.85$; Fig. 2b, middle), a pattern that was consistent across individual participants (Fig. 2b, right). To confirm that this effect was not specific to the choice of dimensionality reduction method, we repeated the analysis using participation ratio—a common alternative measure—which yielded the same result (Supplementary Fig. 1). In summary, this set of findings demonstrates that human PFC population activity is more complex (that is, higher dimensional) than the population activity recorded from M1.

Identification of a low-dimensional CS within the human PFC during context-guided behavior

Having established the high-dimensional nature of human prefrontal population activity, we then investigated whether a low-dimensional subspace within PFC could selectively capture the action information that is relayed downstream to M1. We first characterized PFC–M1 interactions by assessing how well variability in the M1 population activity could be predicted from the PFC population activity, using reduced rank regression (RRR)⁴³ (Fig. 3a and Methods). We then identified the specific dimensions mediating this communication between the two cortical regions^{23,24}, which we termed the PFC–M1 CS. If the communication between the two regions occurs within a subset of the PFC population activity, then a low-dimensional subspace should achieve the same predictive performance as the full regression model (Methods). To test this hypothesis, we constructed a CS for each participant independently. To minimize bias and account for variability in electrode coverage and trial count across participants, we repeatedly

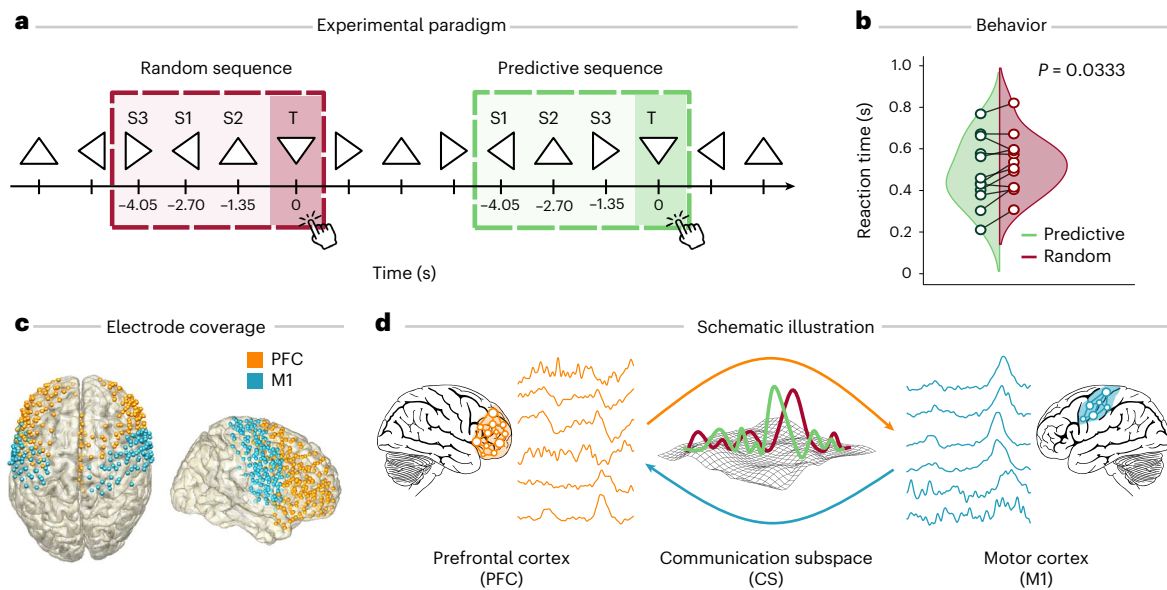


Fig. 1 | Experimental design, electrode placement, behavior and key hypothesis. **a**, Experimental design: participants performed a visual target detection task that featured random and predicted targets (150-ms stimulus presentation, 1.2-s interstimulus intervals). Random targets (red box) were preceded by stimuli that were not presented in a specific order, whereas predicted targets (green box) were signaled by a preceding sequence (clockwise rotating triangles: S1, S2, S3). Targets (T) were presented throughout the presentation blocks and required a button press in response to the target presentation. **b**, Behavioral difference in RTs: group-level distributions of RTs for predictive (green) and random (red) targets, with faster RTs for predictive context ($n = 12$; $P = 0.0333$, $t_{11} = -2.43$, $d = -0.70$; two-sided paired t -test).

c, Electrode coverage in PFC and M1: subdural grid bipolar contacts across 12 participants normalized to an MNI template brain. Left: top view; right: lateral view with electrodes from the left hemisphere mirrored and superimposed onto the right hemisphere for visualization. Electrode contacts are shown over the PFC (orange, 885 contacts) and motor cortex (blue: M1; 316 contacts). **d**, Schematic illustration of the key hypothesis: context-relevant information (green: predictive; red: random) may be encoded within a subspace of high-dimensional PFC population activity. We hypothesized that high-dimensional activity in PFC is projected into a low-dimensional CS that allows the selective relay of only task-relevant information to M1, to effectively guide action execution.

subsampled a fixed number of channels ($n = 15$; Methods) from both regions and averaged trials within each context condition (predictive or random). As hypothesized, the identified PFC–M1 subspace captured the interactions between the human PFC and M1 within a subset of dimensions (PFC–M1, purple) of the total neural population space (full model, gray; Wilcoxon signed-rank test; $P = 0.0010$; Fig. 3b, left). To quantify how the interaction between PFC and M1 differs from interactions within PFC, we also constructed a PFC–PFC subspace, where PFC was split randomly into two subsets, with one predicting the other. For a representative single participant (Fig. 3b, right), six dimensions sufficed to match the predictive performance (R^2) of the full regression model ($R^2 = 0.58 \pm 0.16$; computed with 15 dimensions in this case) for PFC–M1 communication. By contrast, the PFC–PFC subspace required eight dimensions to achieve similar performance. Adding more dimensions to the reduced model did not improve the predictive performance (R^2), indicating that communication both within and across cortical regions occur through a low-dimensional subspace. At the group level, significantly fewer dimensions were needed to capture the PFC–M1 interactions as compared to the interactions within PFC ($P = 0.0010$, $t_{10} = -5.95$, $d = -1.79$; Fig. 3c, left), revealing that the PFC–M1 subspace is lower dimensional than subspaces capturing interactions within PFC. Moreover, the PFC–PFC subspace consistently exhibited higher predictive performance than the PFC–M1 subspace ($P = 0.0023$, $t_{10} = -4.05$, $d = -1.22$; Fig. 3c, right). These results highlight that within-PFC interactions are more complex than the interactions between PFC and M1. To ensure that the lower dimensionality of the PFC–M1 subspace was not inherited trivially from the lower dimensionality of M1, we performed a control analysis in which the same M1 target population was used to compare PFC–M1 interactions with matched within-M1 interactions (Methods). Because both models predicted the same M1 activity, any constraint imposed by M1 affected them equally. Under these matched conditions, the PFC–M1 subspace

remained significantly lower dimensional than the within-M1 subspace ($P = 0.0078$, Wilcoxon signed-rank test; Supplementary Fig. 2), confirming that the low dimensionality of the PFC–M1 CS reflects a selective projection from PFC rather than a constraint imposed by the downstream M1 population. Critically, our findings establish the existence of a low-dimensional CS between PFC and M1 in humans, consisting of a small subset of the overall PFC population activity.

Low-dimensional CS tracks predictive information

After identifying the PFC–M1 CS defined solely on neural population activity patterns, we examined whether this subspace plays a functional role in mediating the context-dependent advantage in behavioral performance. We first extracted the latent PFC activity pattern within this CS by projecting the PFC HFA onto its first dimension (Fig. 4a and Methods). We focused on the first dimension because it consistently captured the largest amount of variance in the CS across participants (Supplementary Fig. 3). This projection reduced the dimensionality of the PFC population activity, while preserving the temporal structure of the neural dynamics that is most predictive of M1 activity⁴⁴. We then assessed the relationship between this low-dimensional representation of PFC and target detection performance. We observed a clear peak in the latent dynamics of the PFC–M1 CS after target onset, suggesting that neural activity within the CS is modulated by target-related processing (Fig. 4a; example participants). Notably, these peaks generally occurred earlier in predictive compared to random contexts for participants who showed a strong behavioral effect (Fig. 4a). To quantify the association between behavior and the latent dynamics of PFC within the PFC–M1 CS, we correlated the peak latency differences between predictive and random peaks of this latent activity with the context-dependent differences in mean RTs across participants. We observed a statistically significant positive correlation ($r_{CS} = 0.79$; $P = 0.0037$, Spearman rank-order correlation; Fig. 4b, left), indicating